

TCS3351 Cryptography and Data Security
Tutorial 3
Classical Cryptosystems

1. Distinguish between the following:
 - i) Substitution and transposition cipher
 - ii) Monoalphabetic and polyalphabetic cipher

Answer:

Substitution : Swap values according to a table

Encrypt		Decrypt	
A	X	X	A
B	V	V	B
...	..	..	..
Z	K	K	Z

Transposition : Change position.

Assume every 8 characters, c

- **Reverse the message:** This is the most basic type of transposition.
 - **Plaintext:** HELLO
 - **Ciphertext:** OLLEH
- **Reverse each word:** The order of words is kept, but the letters within each word are reversed.
 - **Plaintext:** A SIMPLE EXAMPLE
 - **Ciphertext:** A ELPMIS ELPMAXE

2. State the maximum number of characters that will be changed in the ciphertext if only a single character is changed in the plaintext, for each of the following ciphers. Briefly explain the reason for your answer.
 - i) Affine
 - ii) Auto-key
 - iii) One-time pad
 - iv) Rotor
3. State the two problems with one-time pad.

4. Encrypt the message “exe” using the following ciphers. Show also how you can get the original message by performing decryption.
 - i) Additive cipher with $Key = 20$.
 - ii) Multiplicative cipher with $Key = 15$.
 - iii) Affine cipher with $Key = (15, 20)$.

5. Encrypt the following message using Playfair encryption technique.

“The key is hidden under the door pad”

Ignore the spaces between the words. Use **GUIDANCE** as the key. Show also how you can get the original message by performing decryption.

6. Encrypt the message “**meet me**” using the Hill cipher with the key $\begin{pmatrix} 9 & 4 \\ 5 & 7 \end{pmatrix}$. Show also how you can get the original message by performing decryption.
7. Encrypt the plaintext message “**the house is being sold tonight**” using the following ciphers. Ignore the space between words.
 - i) Vigenere cipher with key: “**dollars**”
 - ii) Autokey cipher with $Key = 7$.
 Show also how you can get the original message by performing decryption.

8. Identify the plaintext message using the **Rail Fence cipher of depth 2, 3 and 4**. Given ciphertext is

FLYILDODENTPLULFLEPTOSOSIL

9. Assume the encryption key in two-stage block/columnar transposition cipher is

5 1 6 2 4 3 7

- i) Find the decryption key.
- ii) Using the key, decrypt the following encrypted quotation.

**WUEA IXBA PYRS NHSR PSTS BHRT OAFL
STTB AIUS OUHA FHIZ EIOI ITHT ARNZ**

Show also how you can get the ciphertext by performing encryption on the resulting plaintext.

10. In a **Polybius cipher**, each letter is enciphered as two integers. The key is a 5×5 matrix of characters as in a Playfair cipher. The plaintext is the character in the matrix, the ciphertext is the two integers (each between 1 and 5) representing row and column numbers. Encrypt the word “**multimedia**” using the Polybius cipher with the following key:

	1	2	3	4	5
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1	z	q	p	f	e
2	y	r	o	g	d
3	x	s	n	h	c
4	w	t	m	i / j	b
5	v	u	l	k	a

References:

1. Behrouz A. Forouzan, Debdeep Mukhopadhyay, "Cryptography and Network Security", Third Edition, McGraw-Hill, 2015.
2. William Stallings, "Cryptography and Network Security – Principles and Practice", Seventh Edition, Prentice Hall, 2017.